

## Section 6.5 –Equation of a Line Given Slope and Point

MPM1D

### DO IT NOW!

**Instructions:** Determine the equation of the line, in slope y-intercept form, that has a slope of 3 and goes through the point (2, -5)

**Note:** You can write the equation of a line once you know the slope and y-intercept.

$$y = mx + b$$

Slope

y-intercept

**Step 1: State what you know about the line**

**Step 2: Determine the y-intercept of the line**

To do this we can use the equation  $y = mx + b$ , substitute in values for  $m$ ,  $x$  and  $y$  and then solve for the  $b$  value. Use the point on the line that is given for the  $x$  and  $y$  values.

**STEP 3: Write the equation of the line in slope y-intercept form.**

**Note:** When writing the final equation of the line, plug in values for  $m$  and  $b$ , not for  $x$  and  $y$ .

**Example 1:** Find the equation of the line with a slope of  $\frac{1}{2}$  that passes through (1, 5).

**Step 1: State what you know about the line**

**Step 2: Determine the y-intercept of the line**

**STEP 3: Write the equation of the line in slope y-intercept form.**

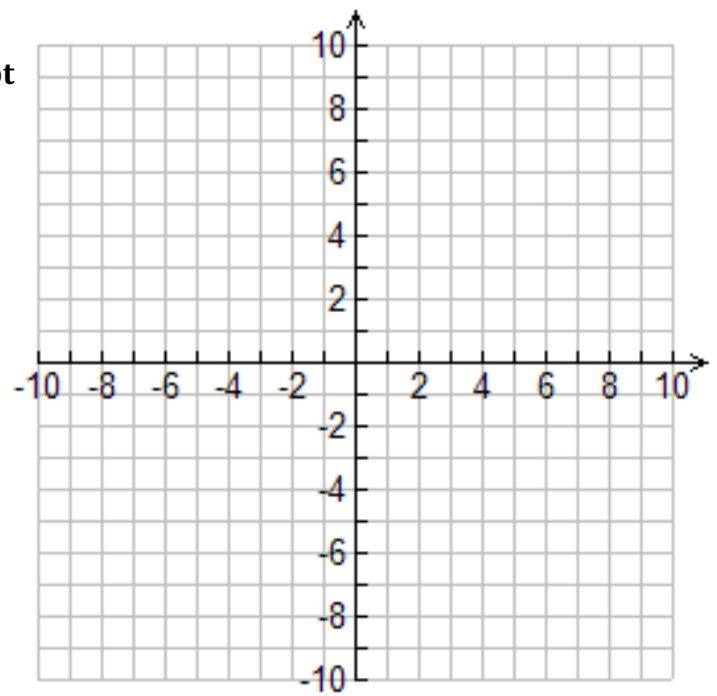
**Example 2:** Find the equation of the line with a slope of 3 and that passes through (0, 2). Then graph the line.

**Step 1: State what you know about the line**

**Step 2: Determine the y-intercept of the line**

**STEP 3: Write the equation of the line in slope y-intercept form.**

**Step 4: Graph the line using the slope and y-intercept**



**Example 3:** Determine the equation of a line that is parallel to the line  $y = -2x - 7$  and passes through the point  $(1, -3)$ .

**Step 1: State what you know about the line**

**Remember:** lines that are parallel have the same slope. They do not have the same y-intercept. You will still have to solve for that.

**Step 2: Determine the y-intercept of the line**

**STEP 3: Write the equation of the line in slope y-intercept form.**

**Example 4:** Determine the equation of a line that is perpendicular to the line  $2x - y + 4 = 0$  and passes through the point  $(-2, 5)$ .

*Hint: to determine the slope you will need to put the equation into  $y=mx+b$  form so that you can see the slope and then take the negative reciprocal.*

**STEP 1: state what you know about the line**

Slope of given line:

Slope of perpendicular line (find negative reciprocal):

Point on the perpendicular line:

**Step 2: Determine the y-intercept of the line** (make sure to use the slope of the perpendicular line)

**STEP 3: Write the equation of the line in slope y-intercept form** (make sure to use the slope of the perpendicular line).

**Consolidation:**

To write the equation of a line you need to know the \_\_\_\_\_ and \_\_\_\_\_.

You can use the slope of a line and a point on the line to calculate the \_\_\_\_\_.

To find the slope of a perpendicular line, find the \_\_\_\_\_.